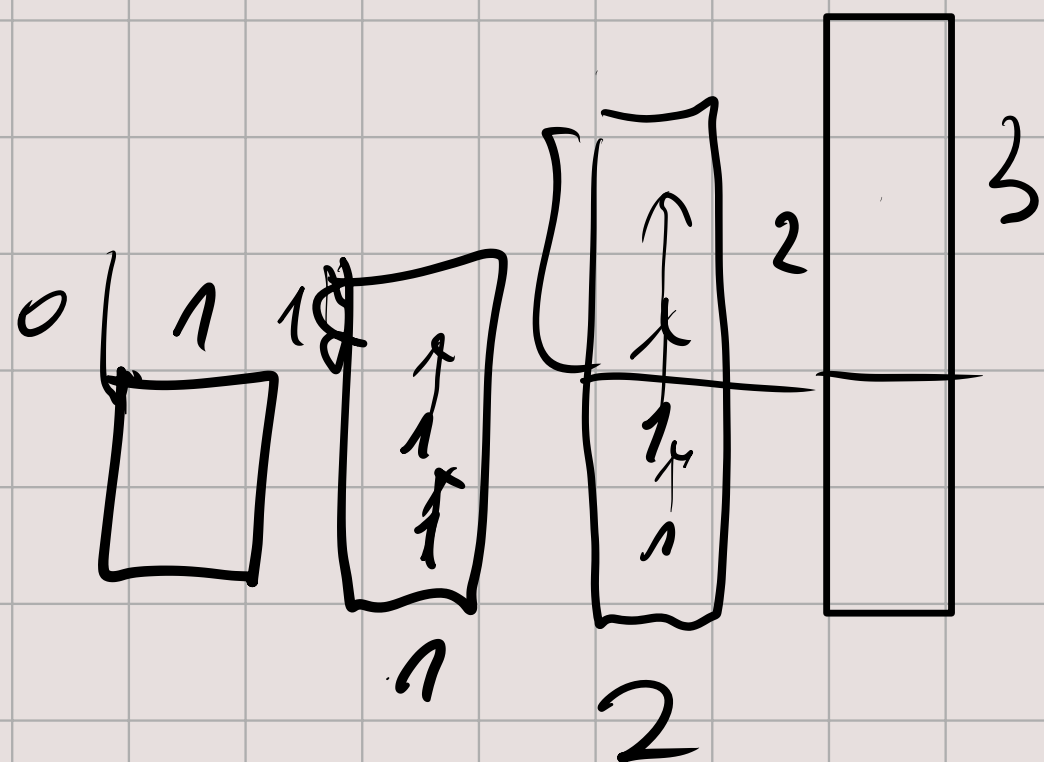


$$\text{Zusatz } \{a_1, a_2, \dots, a_{n+1}\}$$

$$\sum_{k=1}^{n+1} a_k \equiv A \pmod{n}$$

nimm $i \in \{0, 1, \dots, n-1\}$



$$\text{nimm } \sigma_i(\{a_1, a_2, \dots, a_{n+1}\}) =$$

$$\{a_1 + i \pmod{n}, a_2 + i \pmod{n}, \dots, a_{n+1} + i \pmod{n}\}$$

$$\sum_{a \in A} a \equiv \sum_{a \in \sigma_i(A)} a + i \pmod{n}$$

$$\sigma_i(\sigma_j(A)) = \sigma_{j+i}(A)$$

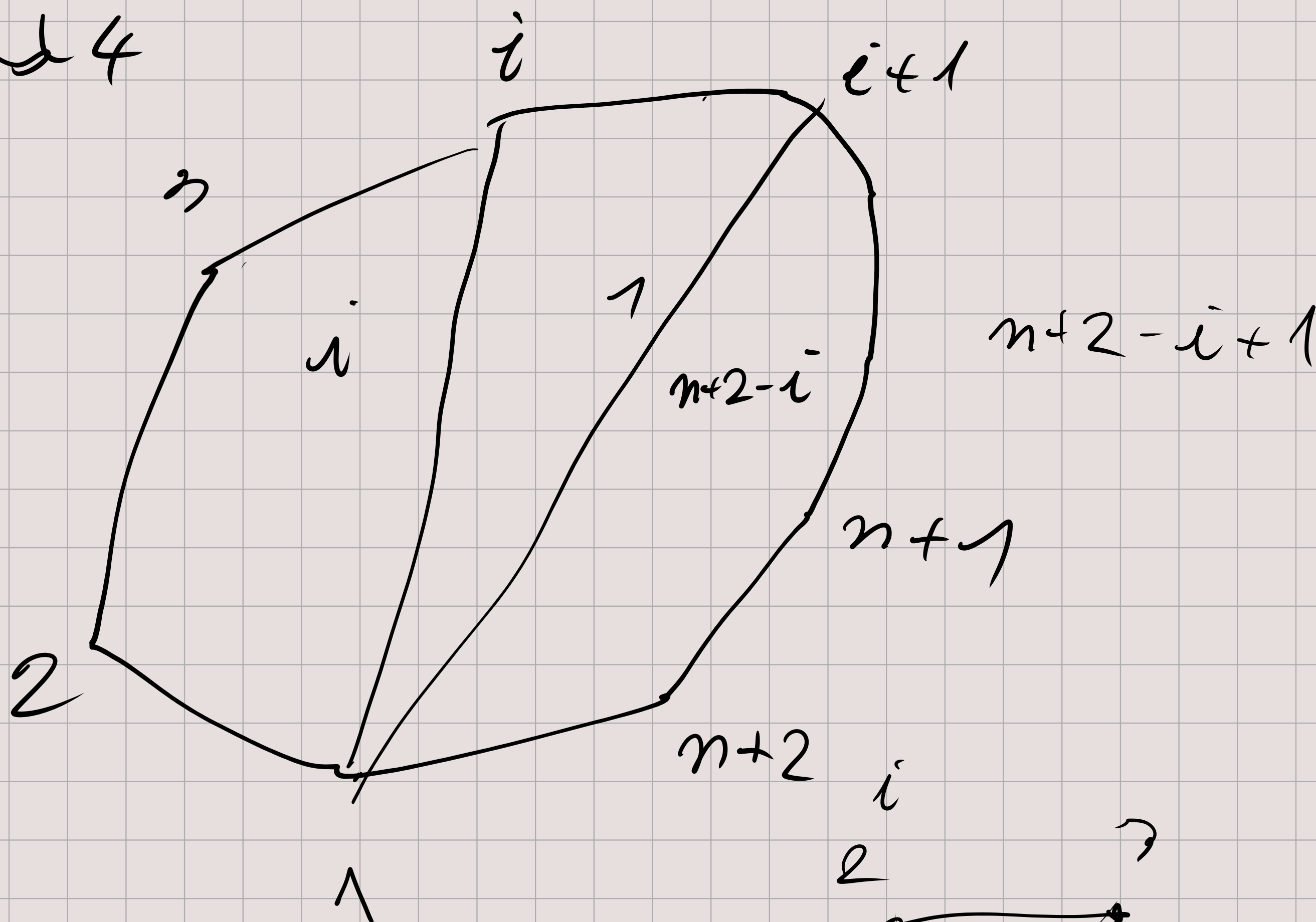
$$\binom{n + (n+1) - 1}{n+1} = \binom{2n}{n+1}$$

$$\{\square, \square, \square\}$$

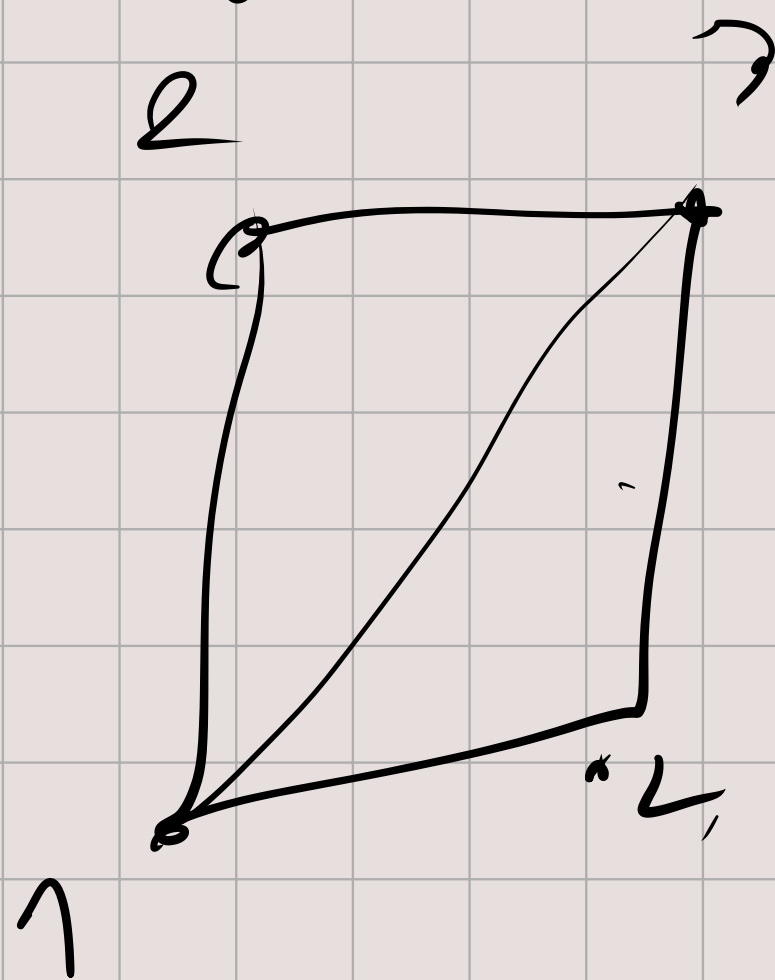
$$\binom{2n}{n+1} = \frac{(2n)!}{(n+1)!(n-1)!} = \frac{(2n)!}{(n!)^2} \cdot \frac{n}{n+1}$$

$$= \binom{2n}{n} \cdot \frac{n}{n+1} = n \cdot C_n$$

Zad 4

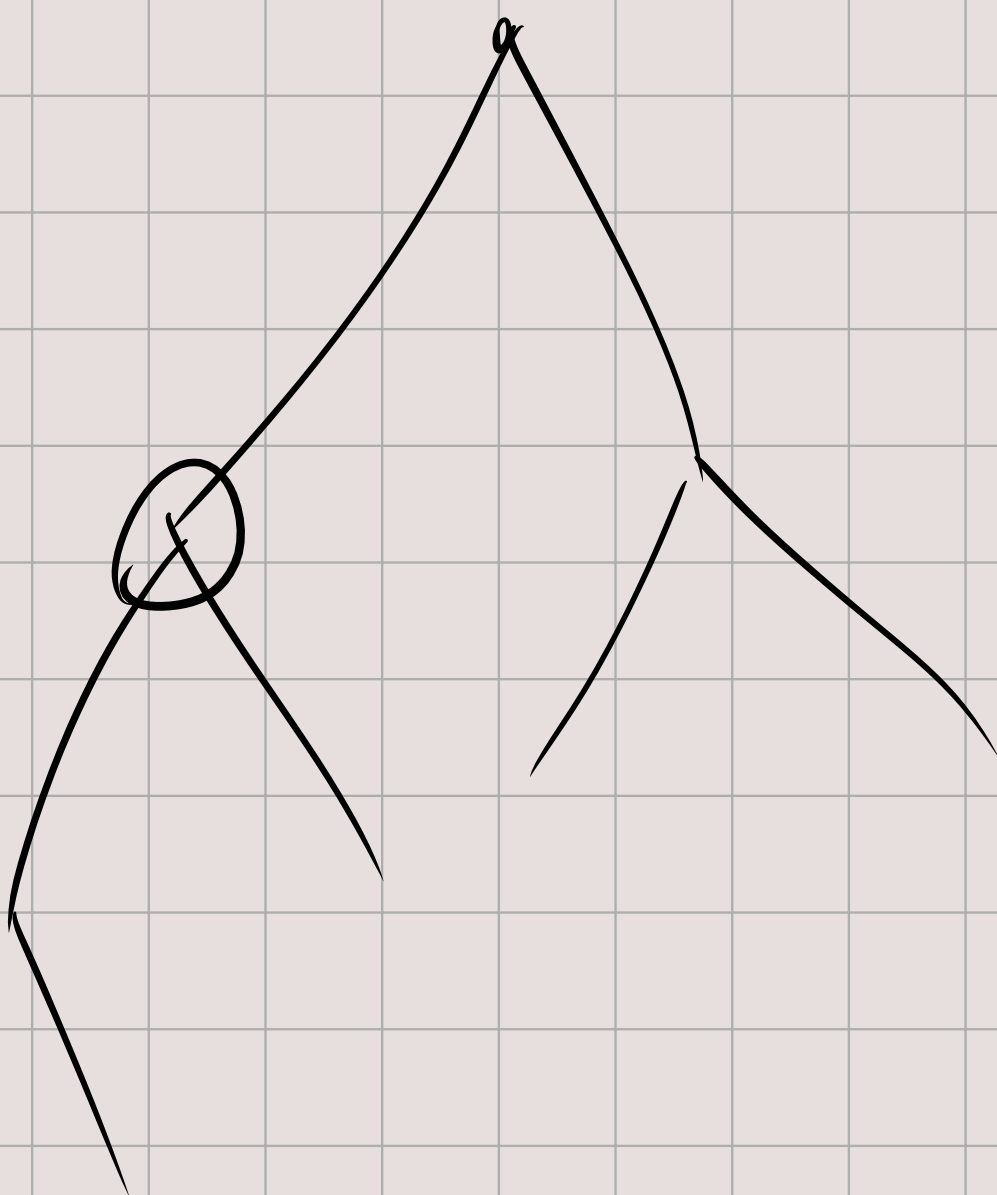
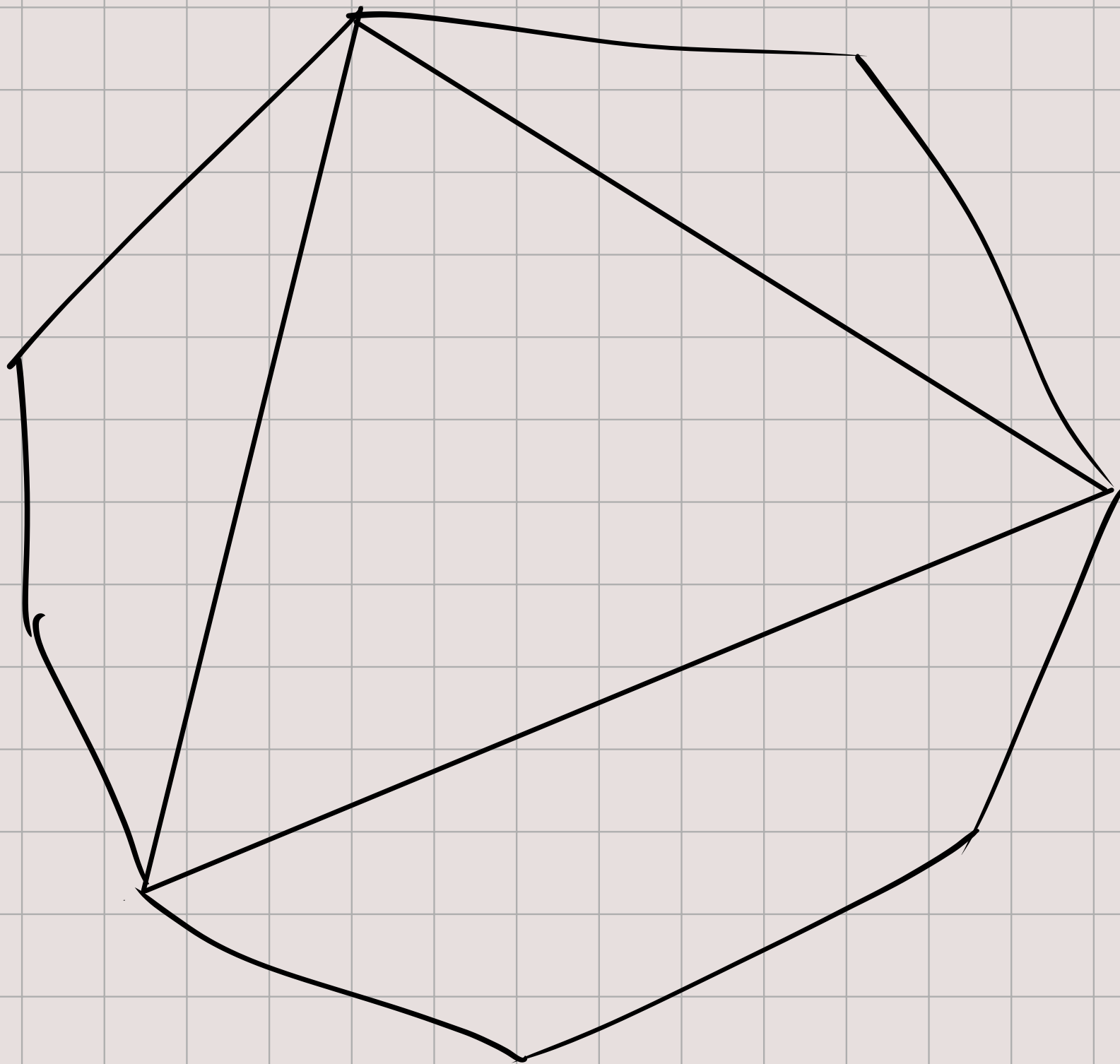


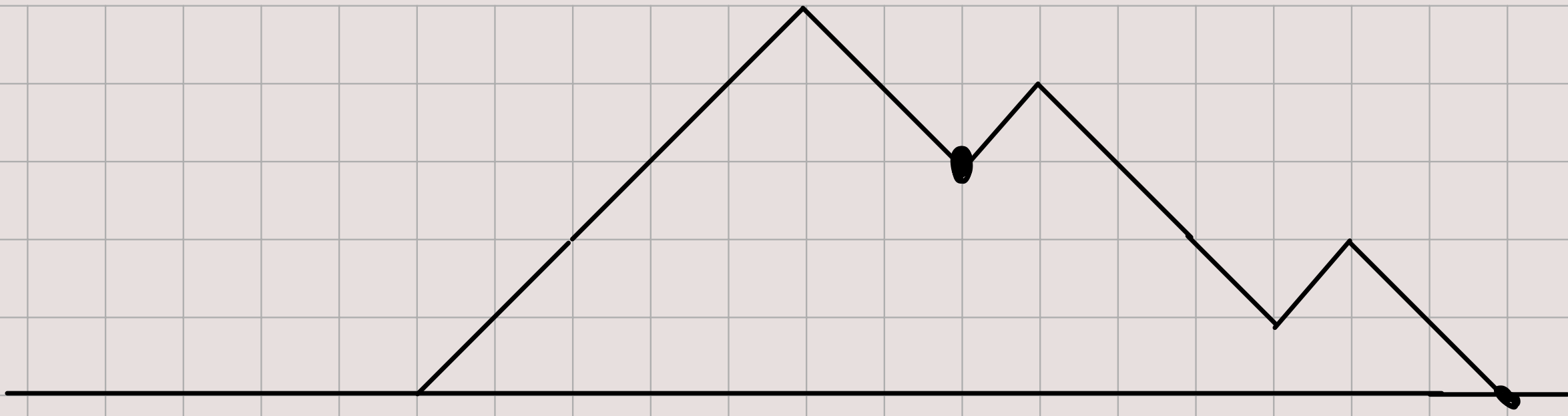
$$\sum_{i=2}^{n+1} C_{i-2} \cdot C_{n+1-i}$$



$$\sum_{i=0}^{n-1}$$

$$C_i \cdot C_{n-1-i} = C_n$$

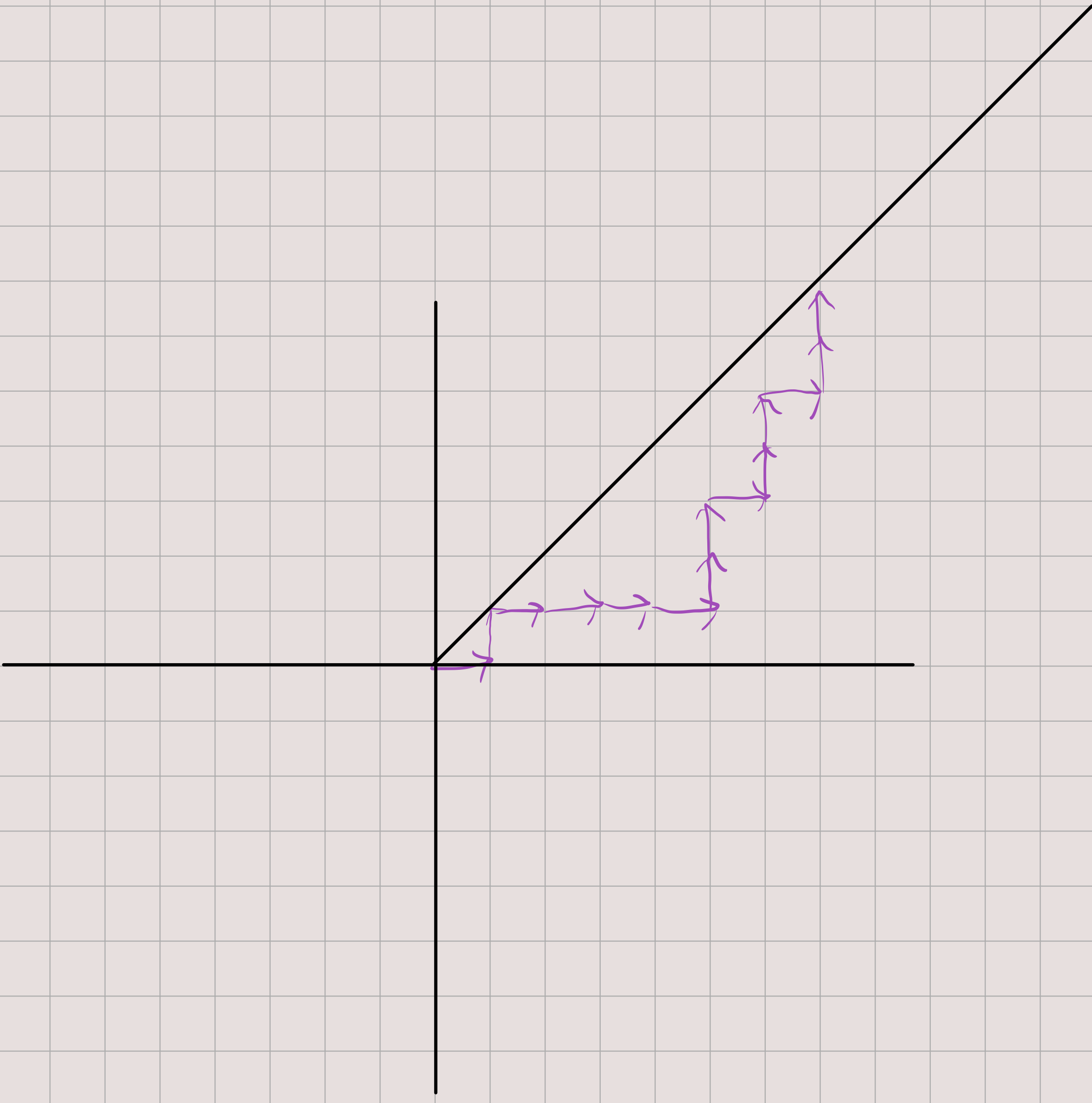
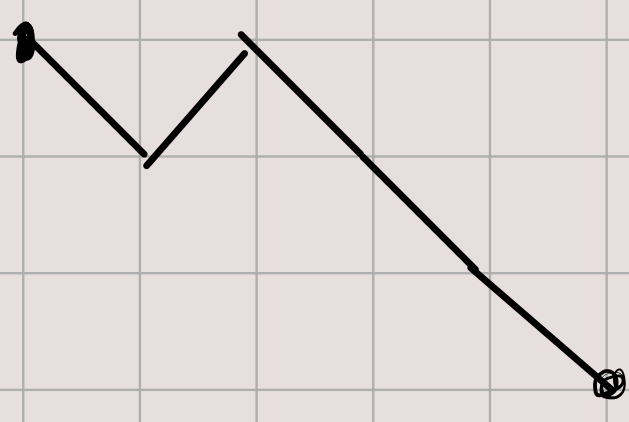




k

$n+1$

$$\sum_{k=1}^{n+1} D_k$$





S_j

$a_1, a_2, a_3, \dots, a_{n-1}$

$$k := \max \{ m \in [n-1] \mid a_m = m \}$$

$\exists!_k$ bo

$\Downarrow$
1

S_k

$\overbrace{a_1 \dots a_n}$

$$\sum_{i=0}^k \binom{n}{i} \binom{n-i}{k-i} = 2^k \binom{n}{k}.$$

$$= \sum_{i=0}^{n-2j} \binom{n}{i} \binom{n-i}{n-2j-i} = 2^{n-2j} \binom{n}{n-2j}$$

||

$$\sum_{i=0}^{n-2j} \binom{n}{i} \binom{n-i}{2j} = 2^{n-2j} \binom{n}{2j}$$

$$\sum_{i=0}^{n-2j} \binom{n}{i} \binom{n-i}{2j} = \sum_{k=0}^n \binom{k}{2i} \binom{n}{k}$$

||

$$\sum_{i=0}^n \binom{n}{i} \binom{n-i}{2j}$$

||

$$\sum_{i=0}^n \binom{n}{n-i} \binom{n-(n-i)}{2j}$$

||

$$\sum_{i=0}^n \binom{n}{i} \binom{l}{2j}$$